# LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

## Quick Revision: Control and Coordination

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### Nervous system

- **Neuron:** dendrite (receives information) → cell body → axon (carries the electrical impulse) → nerve ending.
- **Synapse:** the gap between neurons. At the axon end, chemicals are released that cross the synapse and start an impulse in the next neuron.
- **Reflex action:** a quick, automatic response (hand pulled from a hot object, blinking, sneezing). It is controlled by the spinal cord.
- **Reflex arc:** receptor → sensory neuron → relay neuron in the spinal cord → motor neuron → effector (muscle or gland).

| Part of brain | Main functions |
|---|---|
| Forebrain (cerebrum) | Thinking, intelligence, memory, voluntary actions, sensory areas (sight, hearing, smell), hunger |
| Midbrain | Involuntary actions, e.g. reflex movements of the head and eyes |
| Hindbrain: cerebellum | Posture and balance; precision of voluntary actions (walking in a straight line, picking up a pencil) |
| Hindbrain: medulla | Blood pressure, salivation, vomiting |
| Hindbrain: pons | Involuntary actions, e.g. regulation of breathing |

**Protection:** the brain sits inside the skull in a fluid-filled balloon (shock absorber). The spinal cord is protected by the vertebral column.

### Coordination in plants

- **Touch-me-not (Mimosa):** leaves fold when touched. Cells change shape by changing their water content. No growth is involved.
- **Tropisms** are growth movements towards or away from a stimulus. Phototropism (light): shoot +, root –. Geotropism (gravity): root +, shoot –. Hydrotropism (water) and chemotropism (pollen tube growing towards the ovule).

| Plant hormone | Effect |
|---|---|
| Auxin | Made at the shoot tip; makes cells grow longer; causes bending towards light |
| Gibberellin | Growth of the stem |
| Cytokinin | Promotes cell division (high in fruits and seeds) |
| Abscisic acid | Inhibits growth; causes wilting of leaves |

### Hormones in animals

| Gland | Hormone | Function / problem |
|---|---|---|
| Pituitary | Growth hormone | Growth. Deficiency in childhood → dwarfism |
| Thyroid | Thyroxin | Metabolism of carbohydrates, fats, proteins. Needs iodine (lack → goitre) |
| Pancreas | Insulin | Controls blood sugar. Deficiency → diabetes |
| Adrenal | Adrenaline | Emergencies: faster heartbeat and breathing, blood diverted to muscles |
| Testes | Testosterone | Changes in boys at puberty |
| Ovaries | Oestrogen | Changes in girls at puberty |

| | | |
|---|---|---|
| Hypothalamus | Releasing hormones | Stimulates the pituitary (e.g. growth hormone release) |

**Feedback mechanism:** controls the timing and amount of hormone. If blood sugar rises, the pancreas releases more insulin. As sugar falls, insulin secretion is reduced.

## Quick check

1. Which part of the brain maintains posture and balance?

2. Why is iodised salt advised in our diet?

3. How is the movement of a touch-me-not leaf different from a shoot bending towards light?